Football Player Performance
Analysis Report

A Comprehensive Data Analysis of Forward Players

Metrics & Performance Indicators

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Executive Summary

This report presents a comprehensive data analysis of football player performance metrics (upto and including 2020-09-24), focusing on forward players.

1 Data Preprocessing & Feature Engineering

Prior to analysis, the following preprocessing steps were performed:

- **Missing Values:** All missing values (NA) were imputed using median imputation for numerical features and mode imputation for categorical features.
- **Feature Engineering:** New features were created including:
  - Goals per Match (GPM)
  - Shot Conversion Rate (Goals / Shots)
  - Shooting Accuracy (Shots on Target / Total Shots) $\times$ 100
  - Goals per Shot on Target
- **Column Dropping:** Irrelevant columns (player ID, season indicators, redundant match statistics) were removed.
- **Validation:** All categorical percentage distributions were verified to sum to 100%.

2 Forward Player Analysis

2.1 Key Findings for Forwards

Table 1: Top Forward Player Metrics

Metric	Top Player(s)
Top Goals	Sergio Agüero (180)
Top Assists	Theo Walcott (51)
Fewest Offsides	Brandon Pierrick, Tyler Roberts, Takumi Minamino, Ben Woodburn, Ferran Torres, Steven Bergwijn, Matheus Pereira, Xande Silva, Fábio Silva (0.0)
Top Shot Conversion	Steven Bergwijn (33.3%)
Top Shooting Accuracy	Matej Vydra (54.0%), Gabriel Jesus (54.0%)
Top Goals per Match	Ivan Cavaleiro (0.75), Steven Bergwijn (0.75)
Top Right Foot Scorer	Sergio Agüero (127.0)
Top Left Foot Scorer	Mohamed Salah (63.0)
Top Shots per Match	Sergio Agüero (3.79)
Top Goals per Shot on Target	Ivan Cavaleiro (0.75), Steven Bergwijn (0.75)

2.2 Club and Player Performance Visualizations

Figure 1 displays bar plots for:

- Top 8 clubs by total goals
- Top scorer in each club
- Top 8 clubs by total assists
- Top 8 players by assists

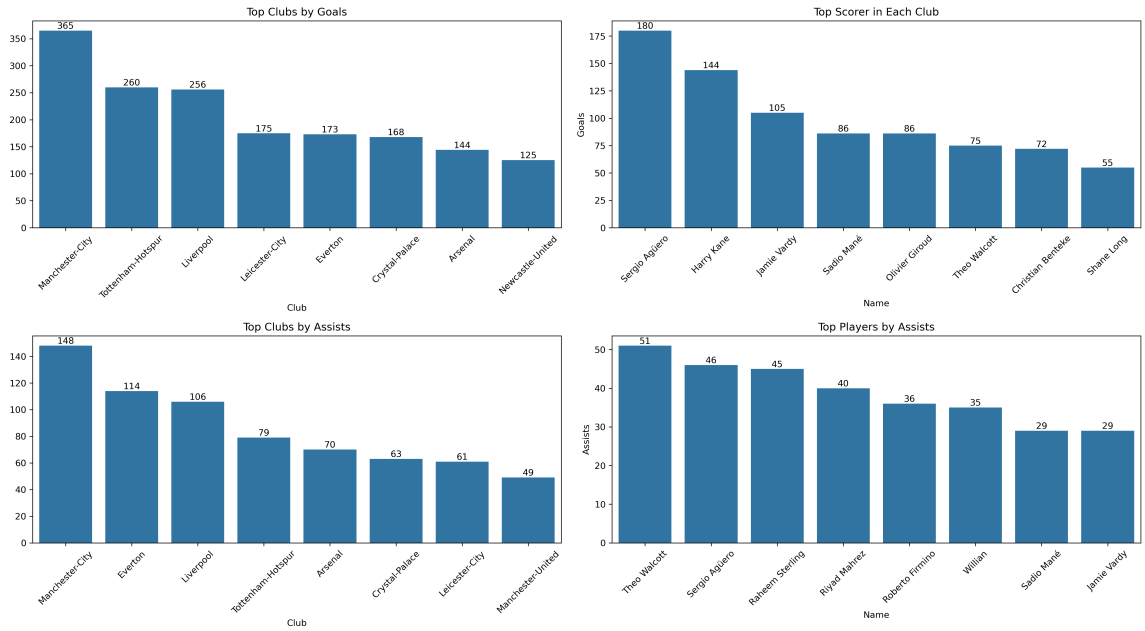

Figure 1: Top clubs by goals and assists, top scorers per club, and top players by assists

2.3 Goals per Match Distribution by Age

Figure 2 shows the relationship between player age and goals per match.

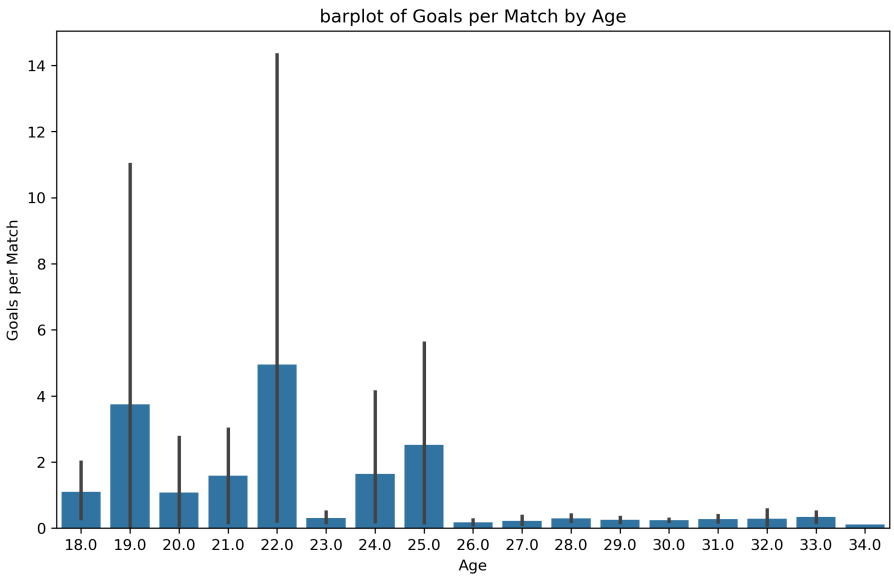

Figure 2: Goals per Match distribution across player ages

The data indicates peak performance in the **21–25 age range**, with the highest goals per match (5.0) at age 23.

2.4 Overall Goal Distribution

Figure 3 presents the histogram of goal counts across all players.

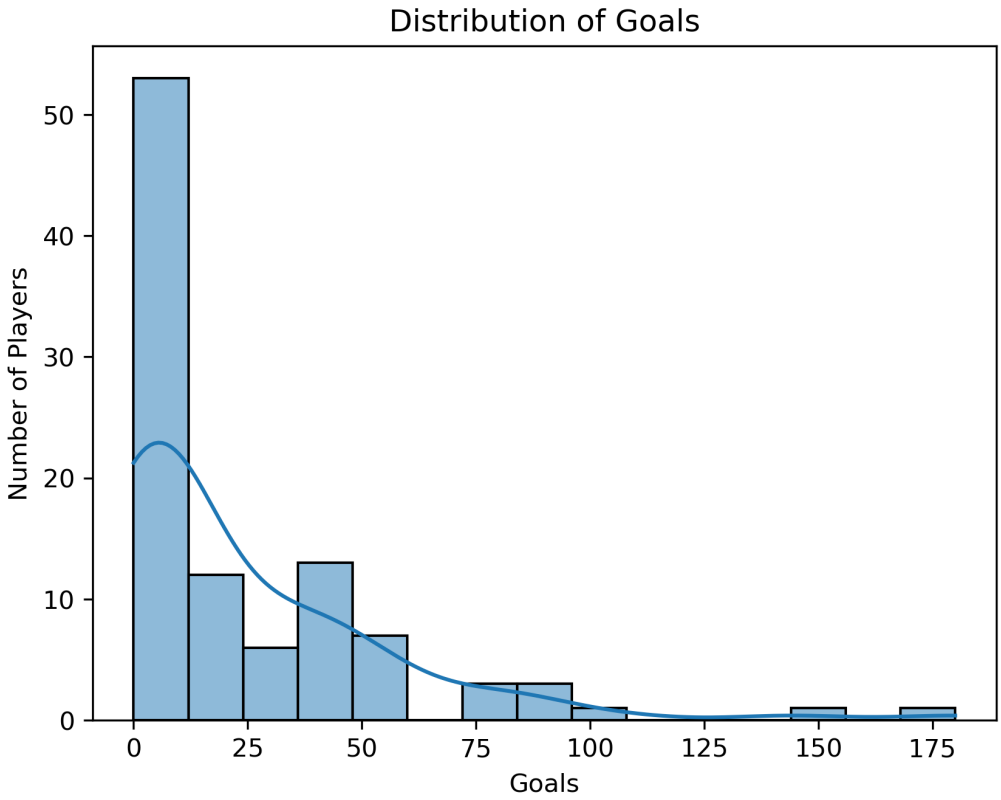

Figure 3: Distribution of goals among all players (X-axis: Goals, Y-axis: Number of Players)

The distribution is **right-skewed**, with most players scoring between 0 and 25 goals, and a long tail extending to 175+ goals for elite scorers.

2.5 Bubble Chart: xG Difference vs Goals and Assists

Figure 4 presents a performance map where:

- **X-axis**: Goals
- **Y-axis**: Assists
- **Bubble size**: Represents xG (expected goals)
- **Color gradient**: xG Difference

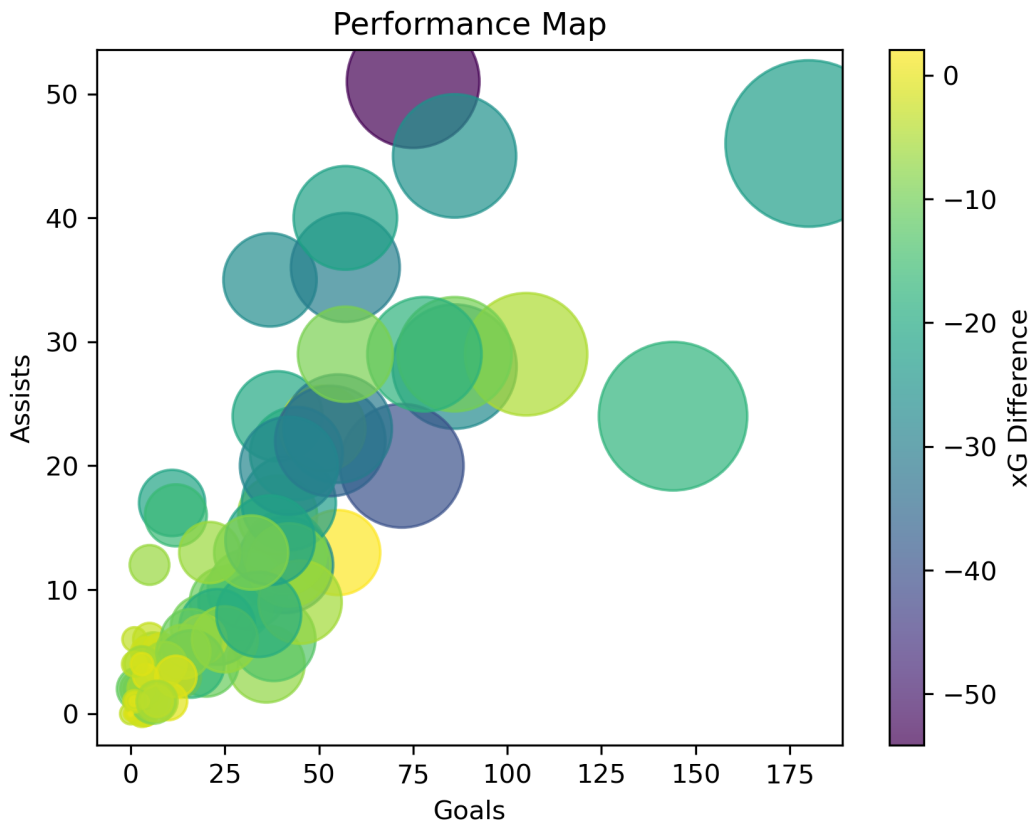

Figure 4: Performance map: Goals vs Assists with bubble size = xG, color = xG Difference

Players with large bubbles and positive xG difference (light colors) **overperform their expected goals**. Those with negative xG difference (dark colors) underperform.

2.6 Age Group Distribution

Figure 5 shows the percentage breakdown of players by age group.

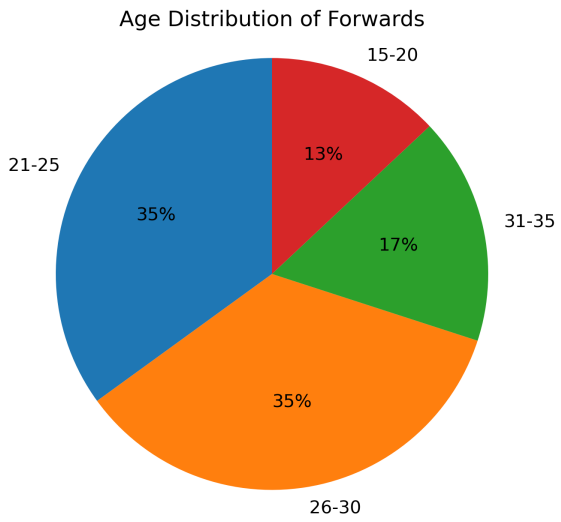

Figure 5: Age group distribution of all players

3 Conclusion & Recommendations

1. **Sergio Agüero** is the standout forward with 180 goals and exceptional shots per match (3.79).
2. **Steven Bergwijn** demonstrates elite efficiency with 33.3% shot conversion and 0.75 goals per match.
3. **Age 23** produces the highest goals per match (5.0), suggesting this as a peak performance age.
4. The goal distribution is highly skewed, with a small number of elite forwards contributing the majority of goals.
5. The xG difference bubble chart identifies overperformers (positive difference) who should be targeted for recruitment.
6. Player age distribution is concentrated in the 21–30 range (70% combined), indicating a healthy mix of prime and developing talent.

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